# The empirical recurrence for OEIS A257355, proved by transfer matrix

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**Abstract**

OEIS A257355 counts arrays of width 4 over $\{0,1\}$ in which every $3 \times 3$ window has constrained SUMS along its rows, its columns and its two diagonals. The entry carries an empirical recurrence of order 66 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a $3 \times 3$ window lies in 3 consecutive array rows, so the count is a walk count on $S = 273$ states.

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## 1 The conjecture

OEIS A257355 is "Number of (n+2)X(2+2) 0..1 arrays with no 3x3 subblock diagonal sum equal to the antidiagonal sum or central row sum less than the central column sum.". It has offset 1 and begins

$$784, 2116, 6724, 20449, 53824, 145924, 451584, 1368900, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 5*a(n-2) +2*a(n-3) +53*a(n-4) +12*a(n-5) -230*a(n-6) -72*a(n-7) -463*a(n-8) -144*a(n-9) +1301*a(n-10) +170*a(n-11) -1891*a(n-12) -1004*a(n-13) +16562*a(n-14) -748*a(n-15) -186*a(n-16) +5100*a(n-17) -2028*a(n-18) +11416*a(n-19) +16030*a(n-20) +35640*a(n-21) -362906*a(n-22) +277568*a(n-23) +114785*a(n-24) +160612*a(n-25) -1868387*a(n-26) +1107206*a(n-27) +814413*a(n-28) +1376008*a(n-29) -5531494*a(n-30) +1143960*a(n-31) +1612833*a(n-32) +5633280*a(n-33) -8273387*a(n-34) -856522*a(n-35) -38781*a(n-36) +10065296*a(n-37) -3435092*a(n-38) -1561924*a(n-39) -2018936*a(n-40) +6596304*a(n-41) +3880456*a(n-42) -174080*a(n-43) -1021152*a(n-44) -1480896*a(n-45) +897952*a(n-46) -1210624*a(n-47) -983712*a(n-48) -1478016*a(n-49) -653440*a(n-50) +296896*a(n-51) +467328*a(n-52) +862208*a(n-53) +413056*a(n-54) +134656*a(n-55) -120576*a(n-56) -237056*a(n-57) -135936*a(n-58) -97792*a(n-59) +15104*a(n-60) +38912*a(n-61) +39936*a(n-62) +18432*a(n-63) -4096*a(n-64) -4096*a(n-65) -4096*a(n-66) for n>68

As of the "Last modified" line on the live entry (Jul 23 2025 15:21:33, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 What is being counted

The array has 4 cells across, with entries in $\{0, 1\}$. A WINDOW is a $3 \times 3$ block of consecutive cells lying inside the array. Every window must satisfy

| | |
|---|---|
| the sum along | |
| the main diagonal | must NOT be equal to the sum along the antidiagonal |
| the central row | must NOT be less than the sum along the central column |

where the main diagonal of a window runs from its top left corner to its bottom right, and its antidiagonal from top right to bottom left.

## 3 The walk

**Lemma 1.** *A* $3\times3$ *window occupies* 3 *consecutive array rows, so every condition above is decided by* 3 *consecutive rows and by nothing else.*

*Proof.* By definition of a window. $\square$

So the state is the window of the last 2 array rows, a row not yet written carried as $\perp$, and a step appends a row and settles every $3 \times 3$ window that row completes. A window needing a row that is $\perp$ does not exist, which is the boundary rule; the walk starts at the all-$\perp$ state, so a walk of $R$ steps is an array of $R$ rows. With $M$ the adjacency matrix and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 273.$$

The walk index $j$ and the entry's own $n$ differ by a constant, read off by matching the published terms rather than assumed. In particular $a$ is $C$-finite.

## 4 The proof

**Lemma 2.** *Let* $q(t) = t^{66} - \sum_i c_i t^{66-i}$ *be the characteristic polynomial of the conjectured recurrence. The residual* $a(n) - \sum_i c_i a(n-i)$ *equals* $\iota^{\top} M^{\,j} q(M)\tau$ *for the corresponding walk index* $j$*, so the recurrence holds from some point on if and only if* $\iota^{\top} M^j q(M)\tau = 0$ *for all large* $j$*, and it is enough to examine* $j < S$*.*

*Proof.* Factor $M^j$ out of $M^{j+66} - \sum_i c_i M^{j+66-i}$. For the bound, $M^S$ is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of $M$; $S$ consecutive zeros force every later one, and the last nonzero $u_j$ pins the threshold exactly. $\square$

**Theorem 3.**

$$a(n) \;=\; 5\,a(n-2)+2\,a(n-3)+53\,a(n-4)+12\,a(n-5)-230\,a(n-6)-72\,a(n-7)-463\,a(n-8)-144\,a(n-9)+1301\,a$$

*for every* $n > 68$*.*

*Proof.* The vector $w = q(M)\tau$ was formed by 66 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 68 + 1$ onwards and the run continued until the working vector was identically zero. $\square$

## 5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order, in the orientation the entry's own name writes, and test a window as soon as its last cell is placed. No state, no transposition and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

## References


[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A257355.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.